

115. What is the mean deviation of first 10 even natural numbers?

- (a) 5
- (b) 5.5
- (c) 10
- (d) 10.5

116. If

$$\sum_{i=1}^{10} x_i = 110 \quad \text{and} \quad \sum_{i=1}^{10} x_i^2 = 1540$$

then what is the variance?

- (a) 22
- (b) 33
- (c) 44
- (d) 55

117. 3-digit numbers are formed using the digits 1, 3, 7 without repetition of digits. A number is randomly selected. What is the probability that the number is divisible by 3?

- (a) 0
- (b) $\frac{1}{3}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{8}$

118. What is the probability that the roots of the equation $x^2 + x + n = 0$ are real, where $n \in \mathbb{N}$ and $n < 4$?

- (a) 0
- (b) $\frac{1}{4}$
- (c) $\frac{1}{3}$
- (d) $\frac{1}{2}$

119. If A and B are two events such that

$$P(\text{not } A) = \frac{7}{10}, \quad P(\text{not } B) = \frac{3}{10} \quad \text{and}$$

$P(A|B) = \frac{3}{14}$, then what is $P(B|A)$ equal to?

- (a) $\frac{11}{14}$
- (b) $\frac{9}{14}$
- (c) $\frac{1}{4}$
- (d) $\frac{1}{2}$

120. Seven white balls and three black balls are randomly placed in a row. What is the probability that no two black balls are placed adjacently?

- (a) $\frac{7}{15}$
- (b) $\frac{8}{15}$
- (c) $\frac{11}{15}$
- (d) $\frac{13}{15}$

SPACE FOR ROUGH WORK